



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2023-24/oprn_SM/687383

Date:

28/11/2023(13:00:09)

Subject:LMI_RELIABILITY IMPROVEMENT MATRIX OF ASH
HANDLING SYSTEM

LMI FOR RELIABILITY IMPROVEMENT MATRIX OF ASH HANDLING SYSTEM has
been prepared inline with OGN reference COS-ISO-00-OIN/OPS/SYST/079 Rev. No.: 01
Sep. 2023

Put up for approval.

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Proposed LMI is reviewed and found in line with OIN. Custodion wil give presentation at DPM on 30.11.2023. File may please be processed further for necessary approval.

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SIMHADRI SUPER THERMAL POWER STATION

LOCATION MANAGEMENT INSTRUCTION

RELIABILITY IMPROVEMENT MATRIX OF ASH HANDLING SYSTEM

LMI/SIMHADRI/
COS-ISO-00-OIN/OPS/SYST/079
Rev. No.: 01 Sep. 2023



ECM ref No.: SMTTP/OPERATION/2023-24/OPR_LMI/687383

Approved for Implementation by:.....

HOP (Simhadri)

Date:

LMI APPROVAL DOCUMENT

LMI Name	Reliability Improvement Matrix of Ash Handling System
LMI No.	LMI/SIMHADRI/OPERATION/OIN/OPS/SYST/079REV-00, Nov-2023
Original / Reference Document	COS-ISO-00-OIN/OPS/SYST/079 Rev. No.: 01 Sep. 2023
Superseded Document	Supersedes: OIN/OPS/SYST/079 Rev. No.: 0 June 2021
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	MATHEW EIPE KOVOOR GM (Maintenance)
Reviewed by	DHURJATI PRASAD PATRA GM(O&M)
Approved by	SANJAY KUMAR SINHA HOP(Simhadri)
Date of Issue	

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RELIABILITY IMPROVEMENT MATRIX OF ASH HANDLING SYSTEM

1.0 OBJECTIVE

For smooth functioning of a thermal power station, Ash Handling Plant (AHP) plays an important role. In order to protect environment, safe disposal of ash in a reliable manner is very important; thus, role of AHP is very crucial. To improve the reliability of AHP system in Simhadri, it is envisaged to formulate a reliability improvement framework. Accordingly, AHP Reliability Improvement Matrix has been prepared, the compliance of which shall be used as basis for evaluation of performance of AHP.

2.0 SUPERSEDED DOCUMENTS- Nil

3.0 SCOPE

AHP Reliability Improvement Matrix of 4x500MW units of Simhadri will help in improving the reliability of AHP equipment and systems. The compliance of this matrix shall be used as basis for evaluation of performance of AHP. Accordingly, weightages have been allocated to respective parameter for evaluation. AHP will be evaluated jointly by Operation and Maintenance every month and exception report along with ratings shall be put up to Head (O&M), HOP, ROS and COS by Head of Operation.

4.0 RELIABILITY IMPROVEMENT MATRIX OF ASH HANDLING SYSTEM

S.No.	Description	UM	Review forum	Frequency	Weightage	Marking Criteria ²¹
A	Generation loss due to Ash Handling System	%	DPM	Monthly	10	If Gen loss is 0, then 10, if up to 0.1%, then 5 otherwise 0
B	Maintaining Healthiness of Fly ash evacuation system				20	
1	Duration of the day (as a percentage of total 24 hours), healthiness of the ash extraction system from ESP to buffer hopper is maintained. [Healthiness in terms of header vacuum as per design, ESP hopper emptiness, cycle time (6Hrs) for evacuation, availability of MHVs & segregating valves, Instrument Air pressure > 5.5ksc at buffer hour top, etc.]	%	DPM	Daily	4	If healthiness > 90% then 4, if > 80% then 2, otherwise 0
2	Duration of the day (as a percentage of total 24 hours), the healthiness of buffer hoppers is maintained (excluding schedule PM hours as per the OGN-OPS-SYST-044). [It includes availability of bag filter assembly, pulsating air system, associated valves, etc.]	%	DPM	Daily	4	If healthiness > 90% then 4, if > 80% then 2, otherwise 0
3	Duration of the day (as a percentage of total 24 hours), the healthiness of vacuum pumps (shut-off vacuum as per design) is maintained.	%	DPM	Daily	3	If healthiness > 90% then 3, otherwise 0

4	Duration of the day (as a percentage of total 24 hours), the healthiness of the ash transportation system from buffer hopper to FA silo/ HCSD system is maintained. [Healthiness in terms of transport air system; buffer hopper emptiness as per design, availability of standby conveying lines, availability of the compressors, associated valves, ALVs, HCSD system, etc.]	%	DPM	Daily	4	If healthiness > 90% then 4, if >80% then 2, otherwise 0
5	Evacuation of APH hoppers (as a percentage of total number of APH hoppers), in each shift.	%	DPM	Daily	3	If emptiness maintained >90%, then 3, otherwise 0
6	Evacuation of Economiser hoppers (as a percentage of total number of Eco hoppers), in each shift.	%	DPM	Daily	2	If emptiness maintained >90%, then 2, otherwise 0
C	Maintaining Healthiness of Bottom ash hopper evacuation system				14	
1	Healthiness of BAH gates (as a percentage of total number of BAH gates) in terms of smooth operation without any leakage.	%	DPM	Daily	4	If healthiness > 100% then 4, , otherwise 0
2	Availability of equipment's below BAH (Hydro ejectors/ SCCs/Screw conveyors/Clinker grinders, etc.) up to ash slurry sump. [In case no generation loss due to above mentioned equipment's, availability is considered as 100%]	%	DPM	Daily	4	If Gen loss is 0, then 4, if up to 0.1%, then 2, otherwise 0
3	Availability of all pumps in Ash handling system (Ash slurry pumps, Pumps in AWP/AWRS etc.) in healthy condition. [In case no generation loss due to the pumps, availability is considered as 100%]	%	DPM	Daily	3	If Gen loss is 0, then 3, if up to 0.1%, then 1, otherwise 0
4	Availability of BA slurry disposal pipelines (including cast basalt lines). [In case no generation loss due to the pipelines, availability is considered as 100%]	%	DPM	Daily	3	If Gen loss is 0, then 3, if up to 0.1%, then 1, otherwise 0
D	System Compliance				10	
Compliance to last 05 HOD meet Best Practices recommendations till date Annex.-01						
1	CHECKING OF VACUUM PUMPS SHUT OFF VACUUM 450 MM HG ONCE IN A DAY	Nos.	Depart mental review	Monthly	2	In case recommendation is compiled then 2, otherwise for each non-compliance 0.5 marks will be deducted

2	Replacing MS slurry lines with Cast basalt pipes as per plan	%	Departmental review	Monthly	2	In case recommendation is compiled then 2, otherwise for each non-compliance 0.5 marks will be deducted
3	Installation of lift for Ash SILO	%	Departmental review	Quarterly	2	In case recommendation is compiled then 2, otherwise for each non-compliance 0.5 marks will be deducted
4	Logging running hrs of dry fly ash evacuation	%	Departmental review	Monthly	2	In case recommendation is compiled then 2, otherwise for each non-compliance 0.5 marks will be deducted
5	Availability of Fire Safety suit for maintenance in bottom ash area	No.	Departmental review	Monthly	2	In case recommendation is compiled then 1, otherwise for each non-compliance 0.5 marks will be deducted

E	Ash utilization and environmental protection				4	
1	Duration of the day (as a percentage of total 24 hours), the Ash silo / hydrobin unloading system is available in healthy condition. [In case there is no ash evacuation/unloading issues, the equipment's will be considered as available].	%	DPM	Daily	2	If availability > 90% then 2, otherwise 0
2	Duration of the day (as a percentage of total 24 hours), the healthiness in terms of fugitive dust emission is maintained. [It includes no leakage from FA transfer lines, buffer hopper bag filters, silo vent fan, HMDC etc.]	%	DPM	Daily	2	If duration of no instance of leakages > 90% then 2, otherwise 0
F	Maintenance practices followed				20	
1	Execution of scheduled PM / Predictive maintenance	%	SAP	daily	2	If > 90% then 2, otherwise 0
2	Execution of Break down maintenance/ Total maintenance	%	SAP	daily	2	If > 20% then 2, otherwise 0
3	Availability of standby equipment's, wherever provided	%	DPM	Monthly	2	If > 90% then 2, otherwise 0
4	Changeover schedule of equipment's, wherever provided	%	DPM	daily	2	If > 90% then 2, otherwise 0
5	Availability of Service Contract and timely initiation of proposal (if applicable) as per NTPC guidelines for Ash Handling system	Yes/No	Departmental review	monthly	2	Yes then 2, other wise 0

6	Duration of the day (as a percentage of total 24 hours), the Interlock and protection system (e.g. ALI of ESP, Buffer hoppers; Bag filter DP, Clinker grinder, slurry pumps/overflow pumps etc.) is available.	%	DPM	daily	2	If > 95% then 1, otherwise 0
7	Duration of the month (as a percentage of total 30 days), the upkeeping of total ash handling area is maintained	%	DPM	Monthly	2	If > 95% then 1, otherwise 0
8	Initiatives taken for improvement in maintenance [e.g., Spare parts development, RCM implementation, repair & maintenance cost reduction, PIP taken & implemented (based on vendor/ supplier interaction; Comparative study/ feedback from other stations / NTPC BE assessment / Audits etc.) and its implementation	Nos	Departmental review	Monthly	5	If no of instances is 2 then 5, if 1 then 2.5, otherwise 0
G	Safety				15	
1	Compliance of safety deviations related to AHS	%	Departmental review	Monthly	2	If >=80%, then 2 and <80% then 0
2	No safety audit (internal, External, CFST, Regional, PTW audit etc.) point pending for more than 1 month / Award and accolades to AHP on safety / Adherence to NTPC health check up by any of the employees	Yes/No (If Yes, Specify)	Departmental review	Monthly	4	If any 2 of the criteria is fulfilled for the month, then 4, If any 1 of the criteria is fulfilled then 2, otherwise 0
3	TBT conducted for Jobs- (No of work permits / No of TBT)	%	Departmental review	Monthly	2	If >= 95 %, then 2 and <95% then 0
4	Number of Reportable accidents in AHS	Nos	Departmental review	Monthly	3	If zero then 3, otherwise 0
5	Number of Non-Reportable / First aid accidents in AHS	Nos	Departmental review	Monthly	2	If zero then 2, otherwise 0
6	Number of Fire Incidents in AHS	Nos	Departmental review	Monthly	2	If zero then 2, otherwise 0
	Note: For a Fatal accident 50 Marks to be deducted.					
H	Learning and Growth				7	
1	Internal customer satisfaction score	%.	Departmental review	monthly	2	If > 90% then 2, 70 to 90% then 1, otherwise 0
2	Participation (as a percentage of total number of employees in AHS) in competency development program /KT meet/HOD meet/GPI Learning	%	Departmental review	monthly	2	If > 90% then 2, 70 to 90% then 1, otherwise 0
3	Engagement of employees (as a percentage of total number of employees in AHS) in QC, PC, 5S etc./ Awareness of LMI and its	Nos.	Departmental review	monthly	3	If > 90% then 3, 90% to 70% then 1, otherwise 0

	compliance					
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5.0 Review

The HOD OPERATION shall review this document on a 2 yearly basis or as necessary.

6.0 Revision

The LMI shall be revised by HOD OPERATION & shall be approved by HOP

7.0 DISTRIBUTION

The report should be distributed to the following:

1. Head of AHS
2. GM(O&M)
3. Head of Project
4. GM(OS-RHQ)-Respective Region
5. GM (Corporate OS-BoP)

Annexure-1

BEST PRACTICES RECOMMENDED BY OS

CHECKING OF VACUUM PUMPS SHUT OFF VACUUM 450 MM HG ONCE IN A DAY

Background

In Dry Ash Handling system, ESP & APH hoppers ash is evacuated pneumatically to Buffer Hoppers by vacuum pumps. Ash transportation media is air, differential pressure between hoppers & Buffer Hopper is vacuum. To get minimum pick up velocity of ash including its designed density, 100% air must enter through air intake valve/orifice located at the end of vacuum conveying line. Vacuum lines, ash intake-Ts, bends, branch valves, etc. are connected with many couplings and Buffer Hopper has many flange joints. These are leakage prone. More over ash is a very abrasive material and valves like MHV, Branch Segregating valves, ALV ash inlet/discharge/vent/Pressure Equalizing valves are eroded very fast. Then vacuum with air flow at ash pick-up zone reduced and as a result ash evacuation rate reduced.

Practice

Checking of shut off vacuum 450mmHg (within the duration of 2 min) including Buffer Hopper & extraction line once in a day. Accordingly corrective actions are taken. It ensures capacity of vacuum pump (85.8m³/min-Simhadri Stg-1, 67.96 m³/min-Simhadri Stg-1&2, 48.13m³/min-Simhadri Stg-1) and minimum system leakages/passing. For vacuum extraction system, quantity of ash extraction depends on quantity of air entering through air intake valve and system DP (vacuum)

Benefits

1. It ensures healthiness of the system
2. Leakages are detected easily by local checking at the time of testing
3. Every day checking & taking corrective actions, further deterioration stopped
4. Improved system reliability

OPERATION OF VACUUM EXTRACTION SYSTEM IN DESIGNED OPERATING RANGE

Background

In Dry Ash Handling system, ESP & APH hoppers ash is evacuated pneumatically to Buffer Hoppers by vacuum pumps. Ash transportation media is air and vacuum pump is maintaining designed airflow with vacuum, differential pressure, between hoppers & Buffer Hopper. This air flow provides minimum pick-up velocity of ash in conveying line with designed ash-air mixture density. Ash-Air mixture density depends on operating vacuum. It is to be ensured without deviation. With decreasing of operating range, density reduced and ash conveying rate reduced. Normally, line vacuum in Stg-1 is 130 -140 mmHg and in stg-2 is 120 to 150 mmHg, MHV closes at 240 mmHg in stage-1 and 380 mmHg in stage-2 ash handling with max. Ash flow, MHV opens at 220 mmHg in Stg-1 & 280 in Stg-2 after partial purging, and hopper empties at 100 mmHg in Stg-1 & 110 mmHg in Stg-2.

Practice

To ensure smooth evacuation of ash from ESP, evacuation system is operated within operating vacuum range, which is 200-240 mmHg for stage-1 and 380-280 mmHg for Stage-2.

Benefits

1. It ensures healthiness of the system
2. Efficient ash evacuation
3. By corrective actions, further deterioration stopped
4. Improved system reliability

STOPPING OF FA SYSTEM 1-2 HRS PER SHIFT

Background

Dry Ash Handling system is designed with more than 25% higher capacity, removal of eight hours ash in six hours. One set of Buffer Hopper & Vacuum Pump evacuates 14 hoppers in 210MW Units, 18 to 22 nos. (For two branches/pass) or 36 to 42 nos. (For four branches/pass) in 500MW & above units, one by one. Total evacuation time of all hoppers of a pass once is called cycle time. When system runs efficiently, less ash content or partial load, cycle completes in one hour and pressure transport line runs with less than designed pressure. Less pressure operation increased line erosion and not energy efficient

Practice

When ESP hoppers emptiness cycle time is less than one hour, system is made off for 1-2hrs per shift. It ensures healthiness of the system & energy conservation.

Benefits

1. It ensures healthiness of the system
2. Efficient ash evacuation
3. Line & material erosion reduced
4. Improved system reliability

DRAW DOWN MODE BOTTOM ASH HOPPER DE-ASHING ONCE IN 24 HRS

Background

In thermal power plants, ash is generated continuously after firing of coal. It is collected in boiler bottom ash hopper, Economizer hoppers, APH, Duct & ESP hoppers. Bottom ash generated in boiler furnace is either continuously removed by Scraper Chain Conveyor, Screw Feeder or stored in water impounded Bottom Ash Hopper. This water impounded hopper ash is removed in four/six hour's interval depending upon the hopper capacity by means of jet pulsation pumps. Bottom Ash de-ashing is done either in maintain mode, water level maintained, or in draw down mode, make up water stopped and emptied completely. In 1st procedure, no harm to refractory and flame stable. 2nd procedure confirmed complete emptiness of hopper.

Practice

In maintained mode evacuation, no harm to refractory and flame stable. Completion of removal of ash is confirmed by visual inspection of discharge quality & time duration. In pull down mode, complete hopper is emptied and flame through S meter level peep hole is noticed. Draw down mode de ashing is done once in 24hrs especially during high ash content/clinkering regime.

Benefits

1. It ensures healthiness of the system
2. It ensures no ash build-up inside the hopper
3. Improved system reliability
4. Reduced generation loss. Due to inadequate evacuation of Bottom Ash, if it is build-up above Boiler S-panel, unit have to taken under shut down and thus generation loss occurred.

ASH SLURRY PIPELINE THICKNESS SURVEY & ROTATION

Background

In Ash Handling System, ash is disposed in the form of slurry to Dyke or in Hydro-bin. Normally these pipelines are made of MSERW, ACI or Cast Basalt. Ash is a very abrasive material and it erodes the MSERW & ACI pipe lines. Normally, erosion starts from bottom portion of pipe line & high velocity zone according to flow profile (nearer to pumping station, bends and after bends). Due to this erosion, pipeline gets thinner & leakages arisen within few years.

Practice

As per six months running hour basis, thickness survey is done & noted. According to previous report, regular thickness survey is carried out at high erosion zones. As per report, portion of pipeline, where thickness reduced to 3mm to 5mm, is rotated by 120 deg. Same procedure repeated for next rotation thus increased the life of pipeline. Also this thickness survey indicates the tentative time of replacement of particular portion of particular line.

Benefits

1. Ash Slurry pipe lines life increased
2. Leakage of line due to erosion reduced
3. Survey indicates right time & right portion of line rotation or replacement
4. Pipe line management improved
5. Budgeting for material & contracts improved & justified
6. Improved safety & environment protection.

Annexure-2

SN	BEST PRACTICES	RESPONSIBILITY	FREQUENCY OF EVALUATION OF EXCEPTIONS
1	CHECKING OF VACUUM PUMPS SHUTOFF VACUUM 450 MM HG ONCE IN A DAY	OPN	Daily
2	OPERATION OF VACUUM EXTRACTION SYSTEM IN DESIGNED OPERATING RANGE	OPN	Daily
3	STOPPING OF FA SYSTEM 1-2 HRS PER SHIFT	OPN	Daily
4	DRAW DOWN MODE BOTTOM ASH HOPPER DE-ASHING ONCE IN 24 HRS	OPN	Daily
5	ASH SLURRY PIPELINE THICKNESS SURVEY & ROTATION	AHM	QUARTERLY